

Bureau of Indian Standards

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New Delhi - 110002

IS 13252:2010

2010

INFORMATION TECHNOLOGY EQUIPMENT - SAFETY

SPECIFICATION

ICS 35.200

BIS 2010

Indian Standard

1 SCOPE

This standard specifies safety requirements for information technology (IT) equipment including computers, printers, monitors, servers, and networking equipment. It covers protection against electric shock, energy-related hazards, fire, heat, mechanical, and radiation hazards.

2 REFERENCES

- IEC 60950-1 - IT Equipment Safety, Part 1: General Requirements
- IS 14711 (Part 1) - Electromagnetic Compatibility
- ISO/IEC 24700 - Energy Efficiency of IT Equipment

3 DEFINITIONS

3.1 Accessible Part

A part that can be touched by a test finger as specified in this standard. An accessible part is one that is within reach of a person during normal use.

4 PROTECTION AGAINST ELECTRIC SHOCK

4.1 Basic Insulation

Equipment shall be constructed so that accessible parts are not live in normal use. Where basic insulation is used, the equipment shall be Class I or Class II construction.

4.2 Leakage Current Limits

Equipment Type	Earth Leakage mA, max	Touch Current mA, max
Portable IT equipment	0.25	0.25
Stationary IT equipment	0.75	0.50
Server racks	3.50	1.00

5 ENERGY-RELATED HAZARDS

5.1 Power Supply

The equipment shall be designed for the rated voltage of 240V, 50 Hz AC supply as per IS 1293. The equipment shall tolerate voltage fluctuations of 10% of the rated voltage.

6 FIRE HAZARDS

6.1 Material Requirements

Enclosure materials shall have a minimum flammability rating of V-1 as per UL 94. Printed circuit boards shall have a minimum rating of V-0. Materials in the path of potential ignition sources shall be non-combustible.

7 MECHANICAL HAZARDS

Moving parts such as fans, disk drives, and paper feed mechanisms shall be enclosed or guarded to prevent injury during normal operation. Sharp edges on enclosures shall be rounded to a minimum radius of 0.5 mm.

8 TEST METHODS

- Dielectric Strength Test: Apply 1500V AC for 1 minute
- Earth Continuity Test: 25A for 5 seconds, voltage drop 2.5V
- Leakage Current Test: As per IEC 60950-1, Clause 5.1
- Temperature Rise: Surface temperature shall not exceed 70C
- Burn Test: Enclosure material V-1 rating per UL 94